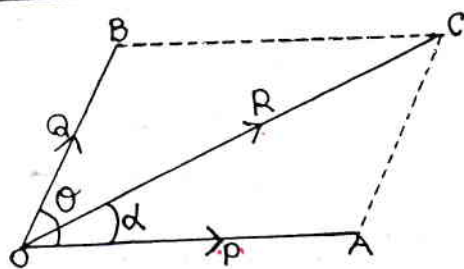


Unit.1 Statics of a particle

The parallelogram of forces



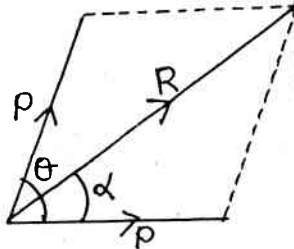
$$R^2 = P^2 + Q^2 + 2PQ \cos \theta \quad (\text{magnitude})$$

$$\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta} \quad (\text{direction})$$

(i) If $P = Q$

$$R = 2P \cos\left(\frac{\theta}{2}\right)$$

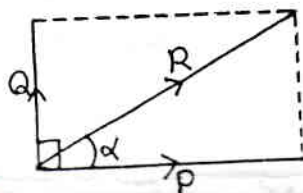
$$\alpha = \frac{\theta}{2}$$



(ii) If $P \perp Q$

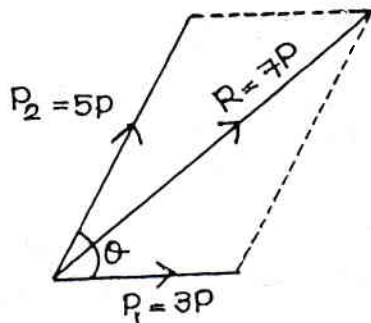
$$R = \sqrt{P^2 + Q^2}$$

$$\tan \alpha = \frac{Q}{P}$$



1 If the resultant of forces $3P, 5P$ is equal to $7P$, find the angle between the forces.

Solution



$$R^2 = P_1^2 + P_2^2 + 2P_1P_2 \cos \theta$$

$$49P^2 = 9P^2 + 25P^2 + 2(3P)(5P) \cos \theta$$

$$30P^2 \cos \theta = 15P^2$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = 60^\circ \leftarrow$$

2 (i) If $P = 10, Q = 10, \theta = 120^\circ$,

find R .

Solution $P = Q = 10$

$$R = 2P \cos\left(\frac{\theta}{2}\right) = 2(10) \cos\left(\frac{120^\circ}{2}\right)$$

$$= 2 \times 10 \times \cos 60^\circ$$

$$R = 2 \times 10 \times \frac{1}{2} = 10 \leftarrow$$

(ii) If $P = 7, Q = 8, \theta = 60^\circ$, find R .

Solution

$$R^2 = P^2 + Q^2 + 2PQ \cos \theta$$

$$= 7^2 + 8^2 + 2(7)(8) \cos 60^\circ$$

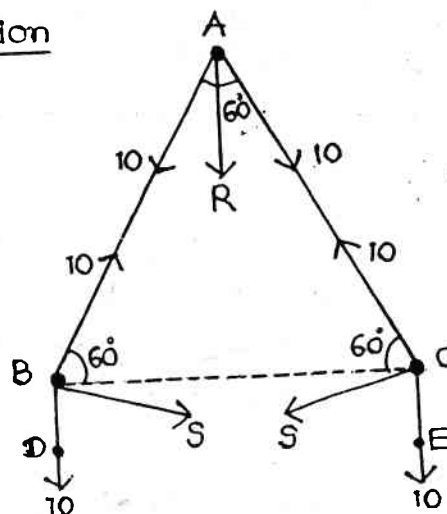
$$= 49 + 64 + 2 \times 7 \times 8 \times \frac{1}{2}$$

$$R^2 = 169$$

$$R = 13 \leftarrow$$

3 Two equal weights of 10 lb are attached to the ends of a thin string which passes over three smooth pegs in a wall arranged in the form of an equilateral triangle with one side horizontal. Find the thrust on each peg.

Solution



On peg A, $R = 2 \times 10 \times \cos\left(\frac{60^\circ}{2}\right)$

$$R = 20 \cos 30^\circ = 20 \times \frac{\sqrt{3}}{2} = 10\sqrt{3} \text{ lb. wt}$$

R bisects $\angle BAC$.

The direction of R is vertically downward.

On peg B and C,

$$S = 2 \times 10 \times \cos\left(\frac{150^\circ}{2}\right) = 20 \cos 75^\circ$$

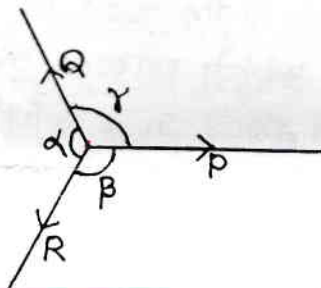
$$S = 20 \times 0.2598 = 5.2 \text{ lb. wt}$$

On peg B, S bisects $\angle ABD$, the direction of S is the angle of depression 15° with BC.

On peg C, S bisects $\angle ACE$, the direction of S is the angle of depression 15° with CB.

Lami's Theorem

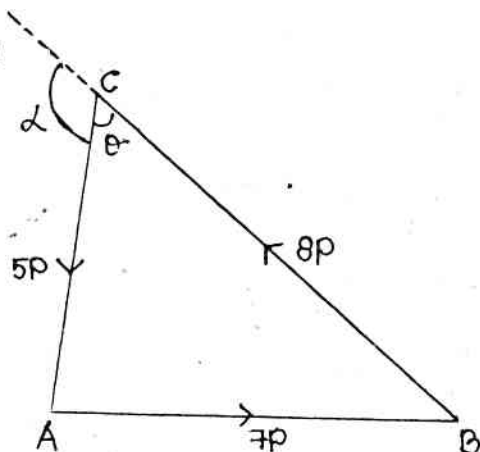
If three forces acting at a point are in equilibrium, each is proportional to the sine of the angle between the other two.



$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}$$

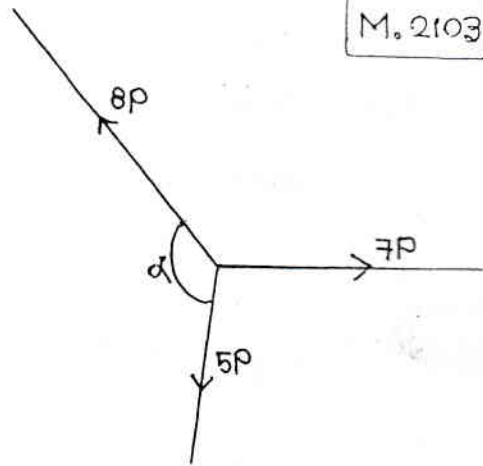
- 4 Forces equal to $7P$, $8P$ and $5P$ acting on a particle are in equilibrium find by drawing and calculation the angle between the forces $8P$ and $5P$.

Solution



QW: 25/5/2022

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In $\triangle ABC$, $AB = 7P$, $BC = 8P$, $CA = 5P$

By the Law of cosine,

$$AB^2 = BC^2 + CA^2 - 2(BC)(CA) \cos \theta$$

$$49P^2 = 64P^2 + 25P^2 - 2(8P)(5P) \cos \theta$$

$$80 \cos \theta = 40$$

$$\cos \theta = \frac{1}{2}$$

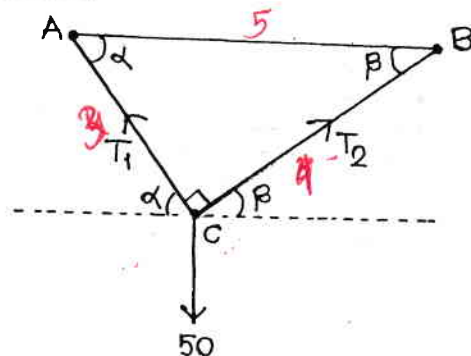
$$\theta = 60^\circ$$

The required angle $= \alpha = 180^\circ - \theta$

$$\therefore \alpha = 180^\circ - 60^\circ = 120^\circ$$

- 5 A particle of mass 50 lb is suspended by two strings 3 ft and 4 ft attached to point at the same level, whose distance apart is 5 ft . Find the tensions in the strings.

Solution



$$AB = 5, AC = 3, BC = 4$$

$$AB^2 = 25, AC^2 + BC^2 = 9 + 16 = 25$$

$$AB^2 = AC^2 + BC^2$$

$$\therefore \angle ACB = 90^\circ$$

By Lami's Theorem,

$$\frac{T_1}{\sin(90^\circ + \beta)} = \frac{T_2}{\sin(90^\circ + \alpha)} = \frac{50}{\sin 90^\circ}$$

$$\frac{T_1}{\cos \beta} = \frac{T_2}{\cos \alpha} = \frac{50}{1}$$

$$\frac{T_1}{\cos \beta} = 50$$

$$T_1 = 50 \cos \beta = 50 \times \frac{4}{5} = 40 \text{ lb. wt} \leftarrow$$

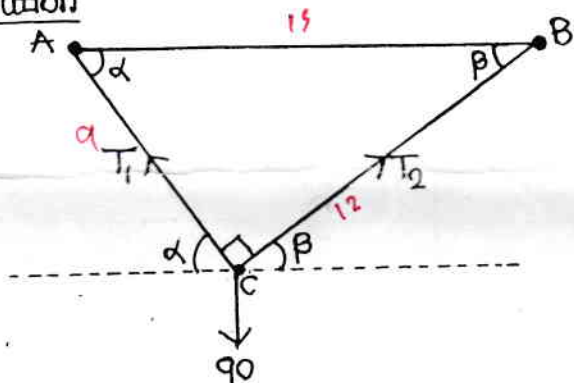
$$\frac{T_2}{\cos \alpha} = 50$$

$$T_2 = 50 \cos \alpha = 50 \times \frac{3}{5} = 30 \text{ lb. wt} \leftarrow$$

sin(90 + \beta) = cos \beta
cos(90 + \alpha) = -sin \alpha

6 A mass of 90 lb is suspended by two light strings of 9 ft and 12 ft from two points at the same level and 15 ft apart. Find the tensions in the string.

Solution



$$AB = 15, AC = 9, BC = 12$$

$$AB^2 = 225, AC^2 + BC^2 = 81 + 144 = 225$$

$$AB^2 = AC^2 + BC^2$$

$$\therefore \angle ACB = 90^\circ$$

By Lami's theorem,

$$\frac{T_1}{\sin(90^\circ + \beta)} = \frac{T_2}{\sin(90^\circ + \alpha)} = \frac{90}{\sin 90^\circ}$$

$$\frac{T_1}{\cos \beta} = \frac{T_2}{\cos \alpha} = \frac{90}{1}$$

$$\frac{T_1}{\cos \beta} = 90$$

$$T_1 = 90 \cos \beta = 90 \times \frac{12}{15} = 72 \text{ lb. wt} \leftarrow$$

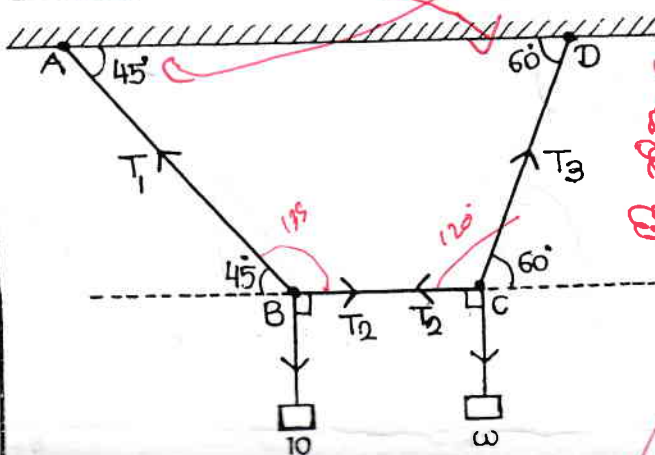
$$T_2 = 90 \cos \alpha = 90 \times \frac{9}{15} = 54 \text{ lb. wt} \leftarrow$$

7 A string ABCD is

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suspended from two points A and D at the same level. A weight of 10 lb is hung from a string knotted to ABCD at B, and a weight w lb. from a string knotted to ABCD at C. If AB and CD make angles of 45° and 60° respectively with the horizontal, and if BC is horizontal, find the tension in the string BC and the weight w .

Solution



Let T_2 = the tension in the string BC

At B, by Lami's theorem,

$$\frac{T_1}{\sin 90^\circ} = \frac{T_2}{\sin 135^\circ} = \frac{10}{\sin 135^\circ}$$

$$\frac{T_1}{1} = \frac{T_2}{\sin 45^\circ} = \frac{10}{\sin 45^\circ}$$

$$\frac{T_2}{\sin 45^\circ} = \frac{10}{\sin 45^\circ}$$

$$T_2 = 10 \text{ lb. wt} \leftarrow$$

At C, by Lami's theorem,

$$\frac{T_2}{\sin 150^\circ} = \frac{w}{\sin 120^\circ} = \frac{T_3}{\sin 90^\circ}$$

$$\frac{10}{\sin 30^\circ} = \frac{w}{\sin 60^\circ} = \frac{T_3}{1}$$

$$\frac{w}{\sin 60^\circ} = \frac{10}{\sin 30^\circ}$$

$$w = \frac{10 \times \sin 60^\circ}{\sin 30^\circ} = 10 \times \frac{\sqrt{3}}{2} \times \frac{2}{1} = 10\sqrt{3} \text{ lb. wt} \leftarrow$$

3 gk & ratios

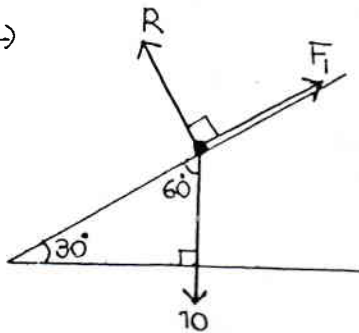
sin 90 = 1
sin 135 = sin 45

8

A particle weighing 10 lb is placed on a smooth plane inclined to the horizontal at an angle of 30° . Find the magnitude of the force (a) acting parallel to the plane (b) acting horizontally, required to keep the particle in equilibrium.

Solution

(a)



Let F_1 = the force acting parallel to the plane

In equilibrium position, by Lami's theorem,

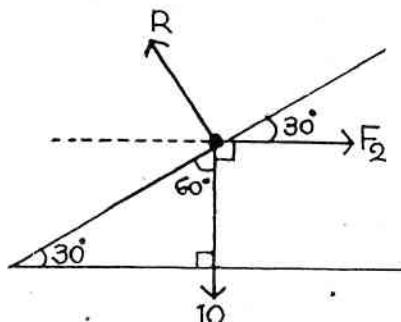
$$\frac{F_1}{\sin 150^\circ} = \frac{R}{\sin 120^\circ} = \frac{10}{\sin 90^\circ}$$

$$\frac{F_1}{\sin 30^\circ} = \frac{R}{\sin 60^\circ} = \frac{10}{1}$$

$$\frac{F_1}{\sin 30^\circ} = 10$$

$$F_1 = 10 \sin 30^\circ = 10 \times \frac{1}{2} = 5 \text{ lb. wt} \leftarrow$$

(b)



Let F_2 = the force acting horizontally

In equilibrium position, by

Lami's theorem,

$$\frac{F_2}{\sin 150^\circ} = \frac{R}{\sin 90^\circ} = \frac{10}{\sin 120^\circ}$$

$$\frac{F_2}{\sin 30^\circ} = \frac{R}{1} = \frac{10}{\sin 60^\circ}$$

$$\frac{F_2}{\sin 30^\circ} = \frac{10}{\sin 60^\circ}$$

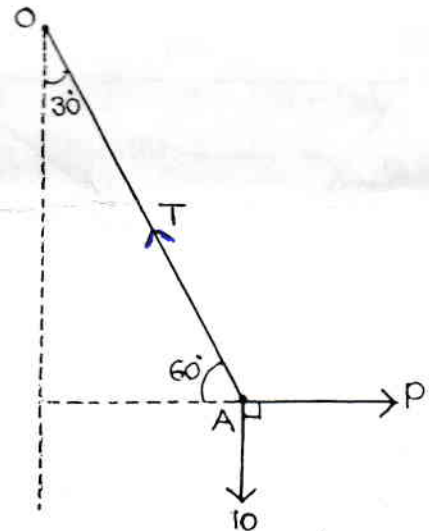
$$F_2 = \frac{10 \times \sin 30^\circ}{\sin 60^\circ} = 10 \times \frac{1}{2} \times \frac{2}{\sqrt{3}}$$

$$F_2 = \frac{10\sqrt{3}}{3} \text{ lb. wt} \leftarrow$$

9

A weight of 10 lb hangs from a fixed-point O by a light inextensible string. It is pulled aside by a horizontal force P lb. wt. and rests in equilibrium with the string inclined at an angle of 30° to the vertical. Find P.

Solution



A is the equilibrium position of the weight.

By Lami's theorem,

$$\frac{T}{\sin 90^\circ} = \frac{P}{\sin 150^\circ} = \frac{10}{\sin 120^\circ}$$

$$\frac{T}{1} = \frac{P}{\sin 60^\circ} = \frac{10}{\sin 60^\circ}$$

$$P = \frac{10 \times \sin 30^\circ}{\sin 60^\circ} = 10 \times \frac{1}{2} \times \frac{2}{\sqrt{3}}$$

$$P = \frac{10}{\sqrt{3}} = 5.77 \text{ lb. wt} \leftarrow$$

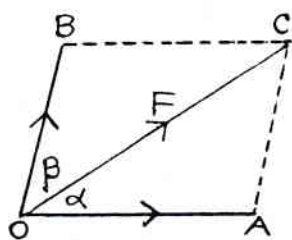
$$\sin(180^\circ - \theta) = \sin \theta$$

$$\sin(90^\circ + \theta) = \cos \theta$$

Resolved a force into components making angles α and β

$$OA = \frac{F \sin \beta}{\sin(\alpha + \beta)}$$

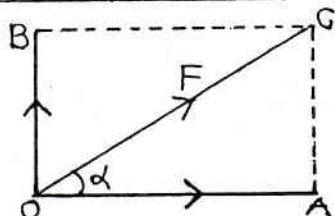
$$OB = \frac{F \sin \alpha}{\sin(\alpha + \beta)}$$



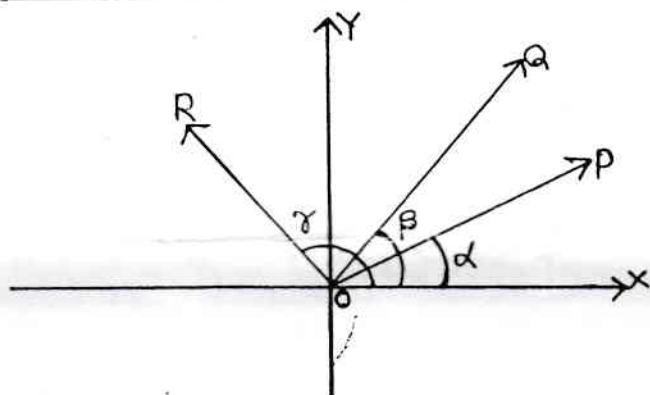
Resolved into two perpendicular components

$$OA = F \cos \alpha$$

$$OB = F \sin \alpha$$



The resultant of any number of forces in one plane acting on a particle



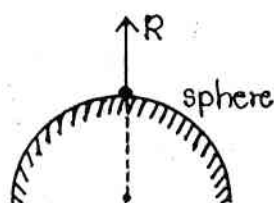
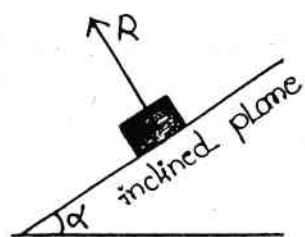
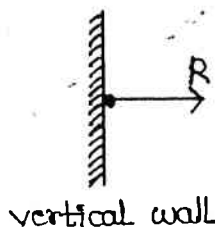
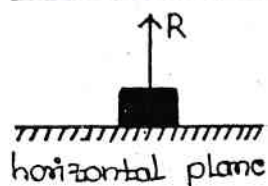
$$X = P \cos \alpha + Q \cos \beta + R \cos \gamma + \dots$$

$$Y = P \sin \alpha + Q \sin \beta + R \sin \gamma + \dots$$

$$\text{Resultant } F = \sqrt{X^2 + Y^2} \quad (\text{magnitude})$$

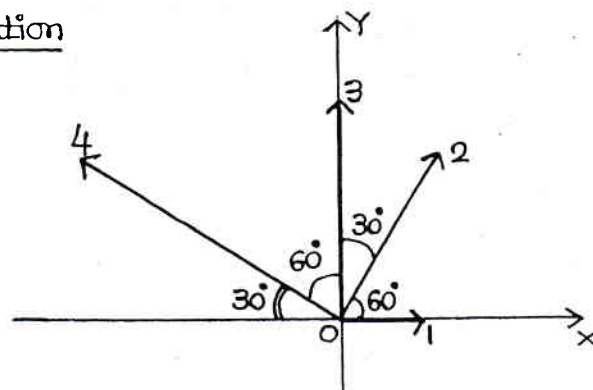
$$\tan \theta = \frac{Y}{X} \quad (\text{direction})$$

Normal reaction R



10 Exam A particle is acted on by forces of 1, 2, 3 and 4 lb. wt, the angles between them being $60^\circ, 30^\circ, 60^\circ$ respectively. Find the magnitude and direction of the resultant.

Solution



The horizontal component of the resultant is $X = 1 \cos 0^\circ + 2 \cos 60^\circ + 3 \cos 90^\circ + 4 \cos 150^\circ$

$$X = 1 + 2 \cos 60^\circ - 4 \cos 30^\circ$$

$$= 1 + 2 \times \frac{1}{2} - 4 \times \frac{\sqrt{3}}{2}$$

$$X = 2 - 2\sqrt{3} = -1.464$$

The vertical component of the resultant is $Y = 1 \sin 0^\circ + 2 \sin 60^\circ + 3 \sin 90^\circ + 4 \sin 150^\circ$

$$Y = 2 \sin 60^\circ + 3 + 4 \sin 30^\circ$$

$$= 2 \times \frac{\sqrt{3}}{2} + 3 + 4 \times \frac{1}{2}$$

$$Y = 5 + \sqrt{3} = 6.732$$

The resultant $= \sqrt{X^2 + Y^2}$

$$= \sqrt{(-1.464)^2 + (6.732)^2}$$

$$= 6.9 \text{ lb. wt} \leftarrow$$

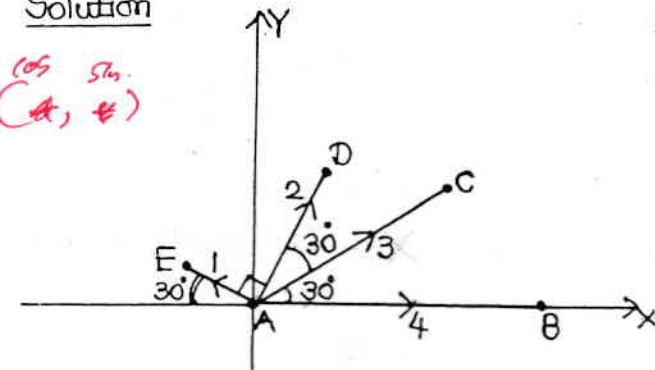
Let θ = angle between resultant and horizontal line

$$\tan \theta = \frac{Y}{X} = \frac{6.732}{-1.464}$$

$$\theta = 102^\circ 16' \text{ with horizontal} \leftarrow$$

- 11 Forces of 4, 3, 2, and 1 lb. wt. act at a point A in directions $\vec{AB}$, $\vec{AC}$, $\vec{AD}$ and $\vec{AE}$ where $\angle BAC = 30^\circ$, $\angle CAD = 30^\circ$, $\angle DAE = 90^\circ$. Find the magnitude of their resultant and the inclination of its direction to AB. Verify by drawing a vector-diagram.

Solution



The horizontal component of the resultant is

$$X = 4 + 3 \cos 30^\circ + 2 \cos 60^\circ - 1 \cos 30^\circ$$

$$= 4 + 3 \times \frac{\sqrt{3}}{2} + 2 \times \frac{1}{2} - 1 \times \frac{\sqrt{3}}{2}$$

$$X = 5 + \sqrt{3}$$

The vertical component of the resultant is

$$Y = 3 \sin 30^\circ + 2 \sin 60^\circ + 1 \sin 30^\circ$$

$$= 3 \times \frac{1}{2} + 2 \times \frac{\sqrt{3}}{2} + 1 \times \frac{1}{2}$$

$$Y = 2 + \sqrt{3}$$

The resultant $= \sqrt{X^2 + Y^2}$

$$= \sqrt{(5 + \sqrt{3})^2 + (2 + \sqrt{3})^2} = 7.7 \text{ lb. wt.}$$

Let θ = the angle between the resultant and AB.

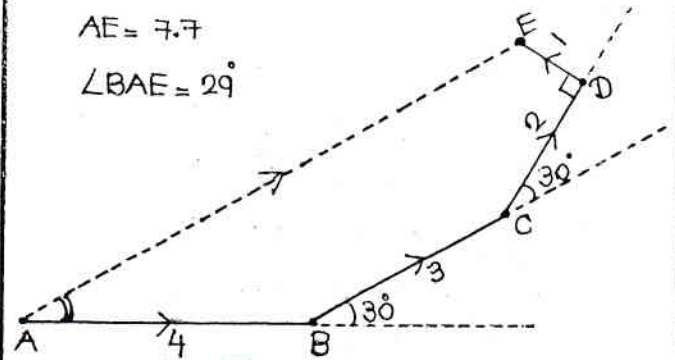
$$\tan \theta = \frac{Y}{X} = \frac{2 + \sqrt{3}}{5 + \sqrt{3}}$$

$$\theta = 29^\circ \text{ with AB}$$

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$$AE = 7.7$$

$$\angle BAE = 29^\circ$$

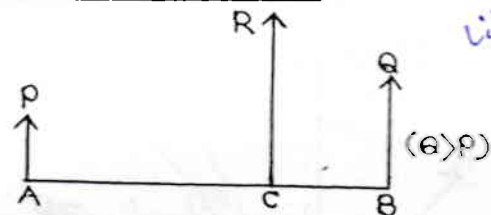


Unit. 2

Statics of a rigid body

Parallel forces - Moments - Couples

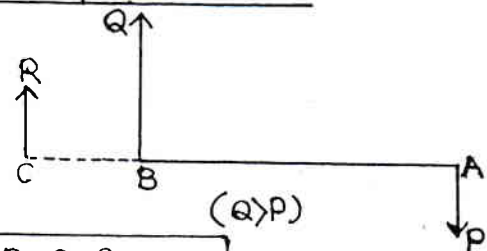
Like parallel forces



$$R = P + Q$$

$$P \times AC = Q \times BC$$

Unlike parallel forces



$$R = Q - P$$

$$P \times AC = Q \times BC$$

Couples



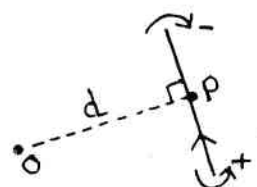
$$R = 0$$

Moment of a force

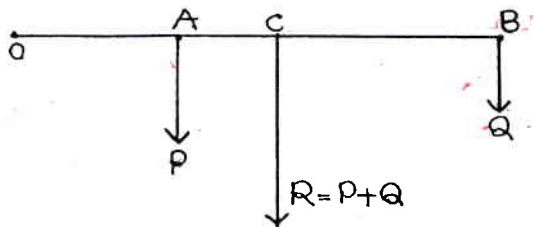
Moment about O = $F \times d$

F = force

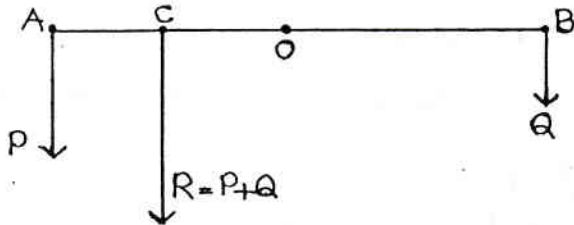
d = perpendicular distance from O



Variation's Theorem



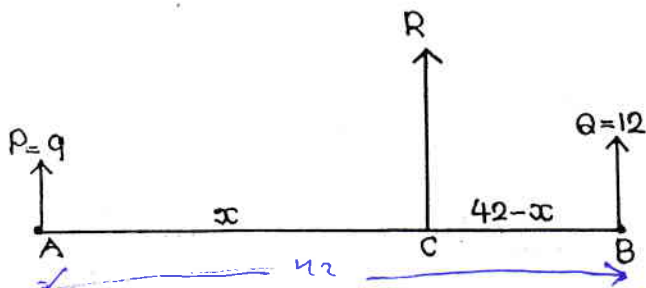
$$R \times OC = P \times OA + Q \times OB$$



$$R \times OC = P \times OA - Q \times OB$$

1. Like parallel forces of 9 and 12 lb. wt. act at points A and B which are 42 in apart. Find the magnitude of their resultant and the point in which it cuts AB. Find also the magnitude and position of the resultant when the forces are unlike.

Solution



If two forces are like,

the resultant $R = 9 + 12 = 21$ lb. wt. ←

$$P \times AC = Q \times BC$$

$$9 \times x = 12(42 - x)$$

$$9x + 12x = 12 \times 42$$

$$21x = 12 \times 42$$

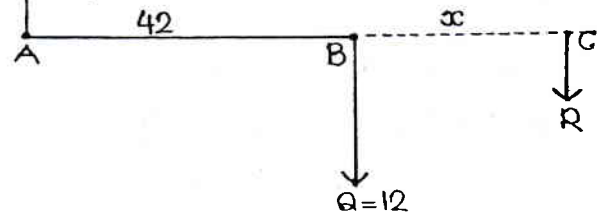
$$x = \frac{12 \times 42}{21} = 24 \text{ in}$$

The resultant cuts AB at a distance 24 in from A. ←

P = 9

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7



If two forces are unlike,

the resultant $R = 12 - 9 = 3$ lb. wt. ←

$$P \times AC = Q \times BC$$

$$9 \times (42 + x) = 12 \times x$$

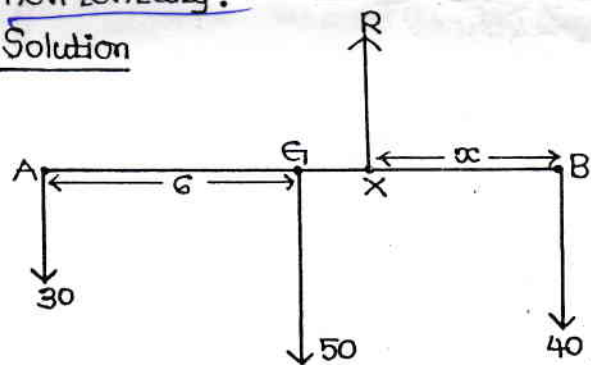
$$9 \times 42 = 12x - 9x$$

$$x = \frac{9 \times 42}{3} = 126 \text{ in}$$

The position of resultant is 126 in from 12 lb. wt force. ←

2. A uniform beam is 12 ft long and weights 50 lb. and weights of 30 and 40 lb are suspended from its ends, at what point must the beam be supported so that it may rest horizontally?

Solution



Let R = supporting force acting on the beam at X

$$R = 30 + 50 + 40 = 120 \text{ lb. wt.}$$

Taking moment about B,

$$R \times BX = 50 \times BG + 30 \times BA$$

$$120x = 50 \times 6 + 30 \times 12$$

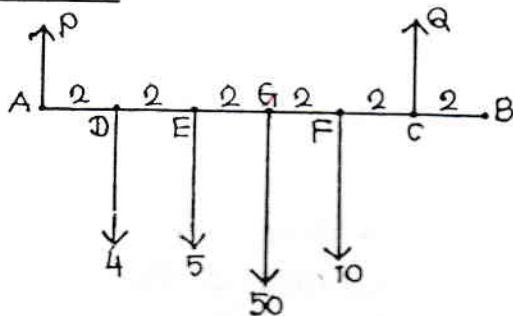
$$120x = 660$$

$$x = \frac{660}{120} = 5.5 \text{ ft.}$$

The point X is 5.5 ft from B. ←

3 A uniform rod AB, of length 12 ft and weighing 50 lb, rests on two supports, one at A and the other 2 ft from B. Weights of 4, 5 and 10 lb. are attached at points 2 ft, 4 ft and 8 ft respectively from A. Find the thrusts on the supports.

Solution



$$P + Q = 4 + 5 + 50 + 10 = 69$$

Taking moments about A,

$$Q \times AC = 4 \times AD + 5 \times AE + 50 \times AG + 10 \times AF$$

$$Q \times 10 = 4 \times 2 + 5 \times 4 + 50 \times 6 + 10 \times 8$$

$$10Q = 408$$

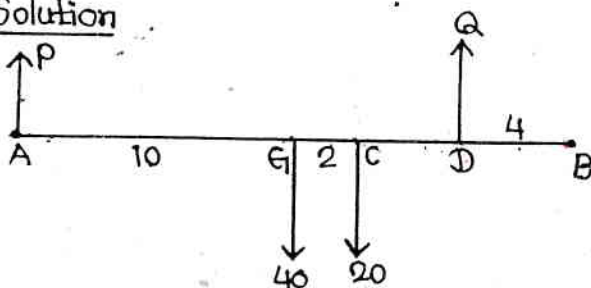
$$Q = \frac{408}{10} = 40.8 \text{ lb. wt.} \leftarrow$$

$$P = 69 - Q = 69 - 40.8$$

$$P = 28.2 \text{ lb. wt.} \leftarrow$$

4 A uniform beam AB, 20 ft long and weighing 40 lb, rests on two supports, one at A and the other 4 ft from B. If a weight of 20 lb is attached to the beam at a point 12 ft from A, find the pressures on the supports.

Solution



$$AB = 20, AG = 10, AC = 12$$

$$P + Q = 40 + 20 = 60$$

Taking moments about A,

$$Q \times AD = 40 \times AG + 20 \times AC$$

$$Q \times 16 = 40 \times 10 + 20 \times 12$$

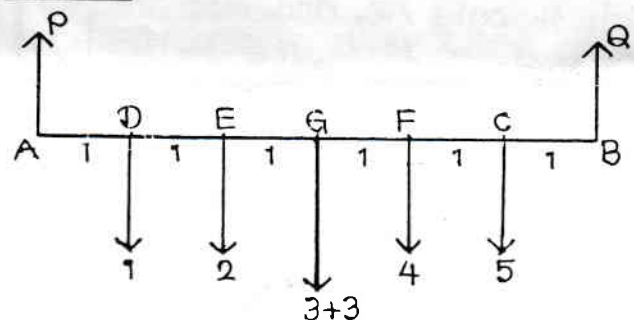
$$16Q = 640$$

$$Q = \frac{640}{16} = 40 \text{ lb. wt.} \leftarrow$$

$$P = 60 - Q = 60 - 40 = 20 \text{ lb. wt.} \leftarrow$$

5 Weights of 1 kg, 2 kg, 3 kg, 4 kg and 5 kg are suspended from a uniform rod AB, 6 m long, weighing 3 kg and supported at its ends, at distances of 1 m, 2 m, 3 m, 4 m, 5 m from A. Find the pressures on the supports.

Solution



Resolving vertically,

$$P + Q = 1 + 2 + 3 + 3 + 4 + 5 = 18$$

Taking moments about B,

$$P \times AB = 1 \times BD + 2 \times BE + 6 \times BG + 4 \times BF + 5 \times BC$$

$$P \times 6 = 1 \times 5 + 2 \times 4 + 6 \times 3 + 4 \times 2 + 5 \times 1$$

$$6P = 44$$

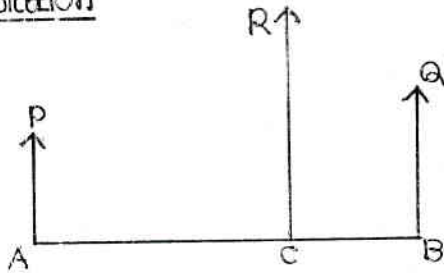
$$P = \frac{44}{6} = \frac{22}{3} \text{ kg} \leftarrow$$

$$Q = 18 - P = 18 - \frac{22}{3} = \frac{32}{3} \text{ kg} \leftarrow$$

6 P and Q are like parallel forces.

If Q is moved parallel to itself, through a distance x , prove that the resultant of P and Q moves through a distance $\frac{Qx}{P+Q}$.

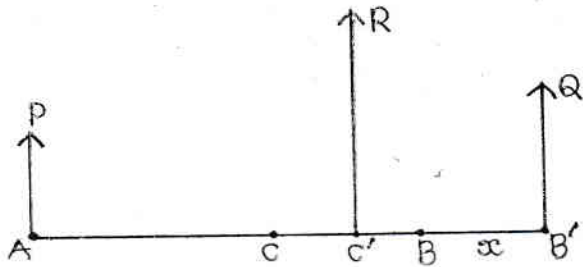
Solution



$$R = P + Q$$

Before Q is moved,

$$P \times AC = Q \times BC$$



Let $BB' = x$.

After Q is moved,

$$P \times AC' = Q \times B'C'$$

$$P(AC + CC') = Q(B'B + B'C')$$

$$P \cdot AC + P \cdot CC' = Q \cdot B'B + Q \cdot B'C'$$

$$= Q \cdot x + Q(BC - CC')$$

$$P \cdot AC + P \cdot CC' = Q \cdot x + Q \cdot BC - Q \cdot CC'$$

$$P \cdot CC' + Q \cdot CC' = Q \cdot x$$

$$CC'(P + Q) = Qx$$

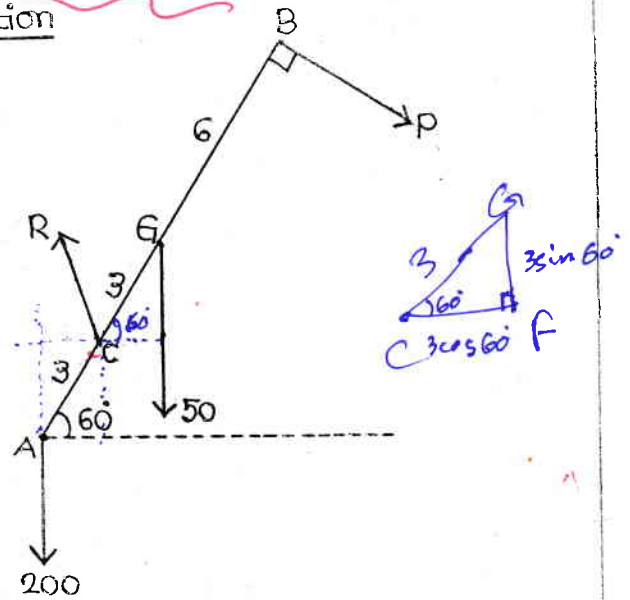
$$CC' = \frac{Qx}{P+Q}$$

7 A uniform rod AB, 12 ft long and weighing 50 lb is pivoted at a point 3 ft from A. A weight of 200 lb is suspended from A. What force applied at B, in a direction perpendicular to the rod, will keep it

in equilibrium with A

below B and AB inclined at 60° to the horizontal?

Solution



$$AB = 12, AG = 6, AC = 3, CG = 3$$

Taking moment about C,

$$P \times BC + 50 \times CG \cos 60^\circ = 200 \times CA \cos 60^\circ$$

$$P \times 9 + 50 \times 3 \times \frac{1}{2} = 200 \times 3 \times \frac{1}{2}$$

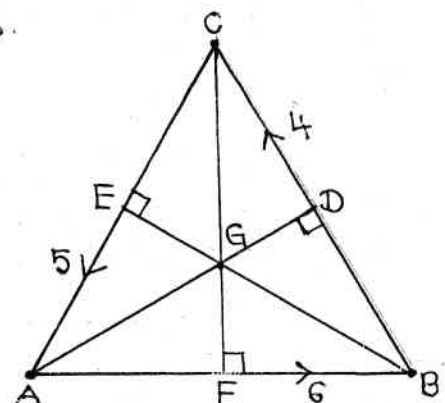
$$9P = 225$$

$$P = \frac{225}{9} = 25 \text{ lb. wt.}$$

The required force is 25 lb. wt.

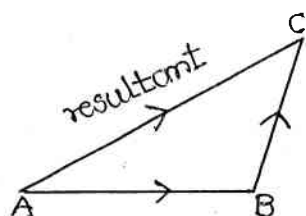
8 Forces of 4, 5 and 6 lb. wt. act along the sides BC, CA, AB of an equilateral triangle ABC of side 2 ft in the directions indicated by the order of the letters. Find the sum of their moments about the point of intersection of the medians of the triangle.

Solution



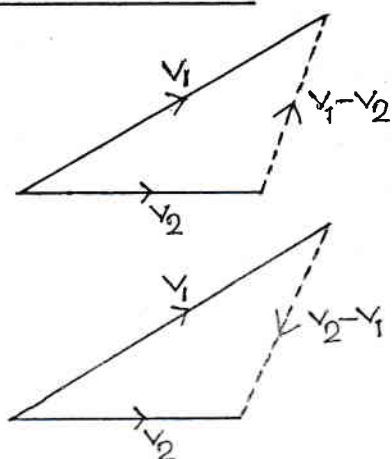
Unit 7 Speed and Velocity

Triangle of velocities



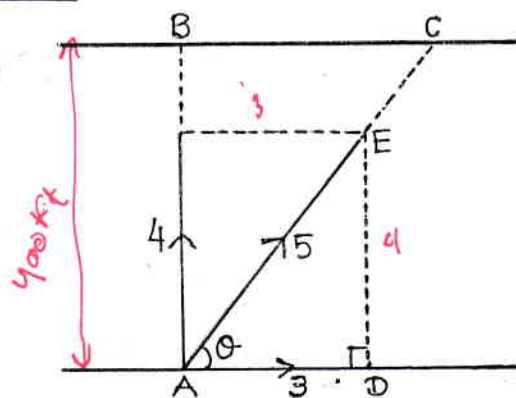
$$\begin{aligned}\vec{AB} + \vec{BC} &= \vec{AC} \\ \vec{BC} &= \vec{AC} - \vec{AB} \\ \vec{CB} &= \vec{AB} - \vec{AC}\end{aligned}$$

Relative velocity



1 A boat is rowed with a velocity of 4 m.p.h straight across a river which is flowing at 3 m.p.h. Find the magnitude and direction of the resultant velocity of the boat. If the breadth of the river is 400 ft, find how far down the river the boat will reach the opposite bank.

Solution



velocity of the boat = 4 m.p.h

velocity of the river = 3 m.p.h

The resultant velocity $v = \sqrt{3^2 + 4^2} = \sqrt{25}$

$$v = 5 \text{ m.p.h} \leftarrow$$

In $\triangle ADE$, $\cos \theta = \frac{3}{5}$

$$\theta = \cos^{-1} \frac{3}{5}$$

The direction of resultant velocity makes $\theta = \cos^{-1} \frac{3}{5}$ with the bank. $\leftarrow$

$$AB = 400 \text{ ft}$$

$$\frac{BC}{AB} = \cot \theta$$

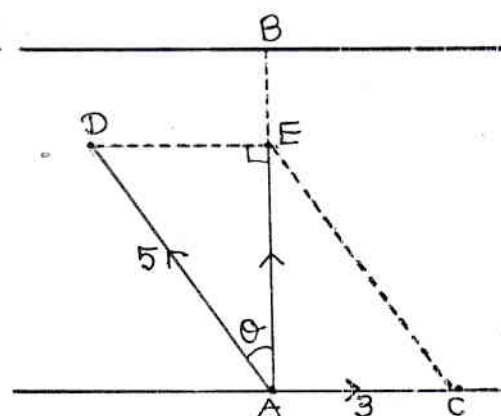
$$\frac{BC}{400} = \frac{3}{4}$$

$$BC = \frac{3}{4} \times 400 = 300 \text{ ft}$$

The boat will be carried downstream a distance of 300 ft. $\leftarrow$

2 A stream is running at 3 m.p.h and its breadth is 440 ft. If a man can row a boat at 5 m.p.h, find the direction in which he must row in order to go straight across the stream, and the time it takes him to cross.

Solution



velocity of stream = 3 m.p.h

velocity of boat = 5 m.p.h

$$\text{In } \triangle ADE, AE^2 = AD^2 - DE^2 = 5^2 - 3^2$$

$$AE^2 = 16$$

$$AE = 4$$

The resultant velocity $v = 4 \text{ m.p.h}$

$$v = 4 \times \frac{5280}{60 \times 60} \text{ ft per sec} = \frac{88}{15} \text{ ft. per. sec}$$

$$s = AB = 440 \text{ ft}$$

The time taken to cross = $t = \frac{s}{v}$

$$t = 440 \times \frac{15}{88} = 75 \text{ sec}$$

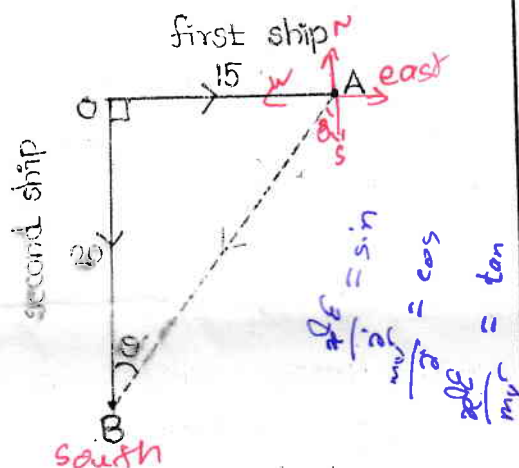
$$\cos \theta = \frac{4}{5}$$

$$\theta = \cos^{-1} \frac{4}{5}$$

The man must row in a direction making an angle $\cos^{-1} \frac{4}{5}$ with AB.

3 A ship is steaming due east at 15 m.p.h and another ship is steaming due south at 20 m.p.h, find the velocity of the second ship relative to the first.

Solution



The velocity of the second ship relative to the first = $\vec{AB}$

In $\triangle AOB$,

$$AB^2 = OA^2 + OB^2 = (15)^2 + (20)^2 = 625$$

$$AB = 25$$

$$\tan \theta = \frac{OA}{OB} = \frac{15}{20} = \frac{3}{4}$$

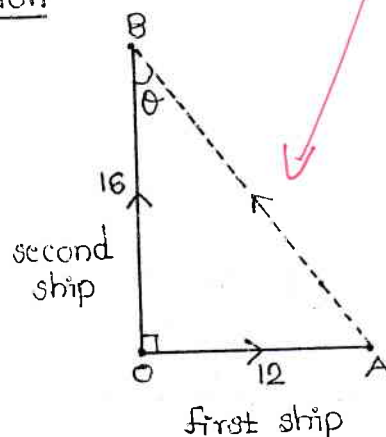
$$\theta = \tan^{-1} \frac{3}{4}$$

The relative velocity is 25 m.p.h, its direction makes an angle $\tan^{-1} \frac{3}{4}$ west of south or S 36.87° W.

4

One ship is sailing due east at 12 m.p.h and another ship is sailing due north at 16 m.p.h, find the velocity of the second ship relative to the first.

Solution



The velocity of the second ship relative to the first = $\vec{AB}$

In $\triangle AOB$,

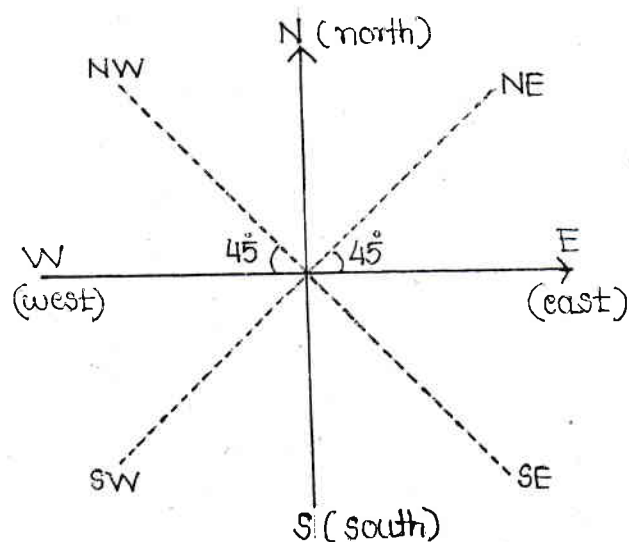
$$AB^2 = OA^2 + OB^2 = (12)^2 + (16)^2 = 400$$

$$AB = 20$$

$$\tan \theta = \frac{OA}{OB} = \frac{12}{16} = \frac{3}{4}$$

$$\theta = \tan^{-1} \frac{3}{4}$$

The relative velocity is 20 m.p.h, its direction makes an angle $\tan^{-1} \frac{3}{4}$ west of north or N 36.87° W.



5 A train is travelling along a horizontal rail at 45 m.p.h, and rain is falling vertically with a velocity of 22 ft per sec. Find the apparent direction and velocity of the rain to a person travelling in the train.

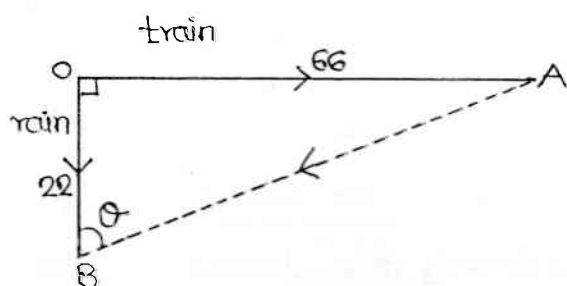
Solution

The velocity of the train = 45 m.p.h

$$= 45 \times \frac{5280}{60 \times 60} \text{ ft per sec.}$$

$$= 66 \text{ ft. per sec.}$$

velocity of the rain = 22 ft. per sec.



The apparent velocity of the rain = $\vec{AB}$

In $\triangle AOB$,

$$AB^2 = OA^2 + OB^2 = (66)^2 + (22)^2 = 4840$$

$$AB = 22\sqrt{10}$$

$$\tan \theta = \frac{OA}{OB} = \frac{66}{22} = 3$$

$$\theta = \tan^{-1} 3$$

The apparent velocity of the rain is $22\sqrt{10}$ ft. per sec, its direction makes an angle $\tan^{-1} 3$ with the velocity of rain.

6 Raindrops are falling through the air with a velocity of 10 ft. per sec. If a north wind blows at 12 m.p.h, find the direction in which the drops appear to fall to a person walking north at 3 m.p.h.

With what velocity

would they hit his umbrella?

Solution

velocity of raindrops = 10 ft. per sec.

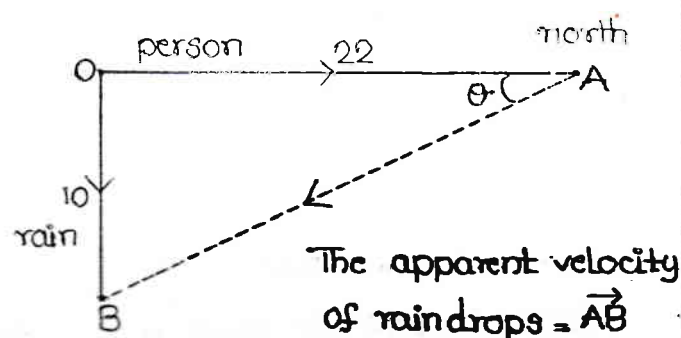
velocity of wind = 12 m.p.h

velocity of person walking = 3 m.p.h

The velocity of person relative to the wind = $12 + 3 = 15$ m.p.h

$$= 15 \times \frac{5280}{60 \times 60} \text{ ft. per sec.}$$

$$= 22 \text{ ft. per sec.}$$



In $\triangle AOB$, $AB^2 = OA^2 + OB^2$

$$AB^2 = (22)^2 + (10)^2 = 584$$

$$AB = 24.17$$

$$\tan \theta = \frac{OB}{OA} = \frac{10}{22} = \frac{5}{11}$$

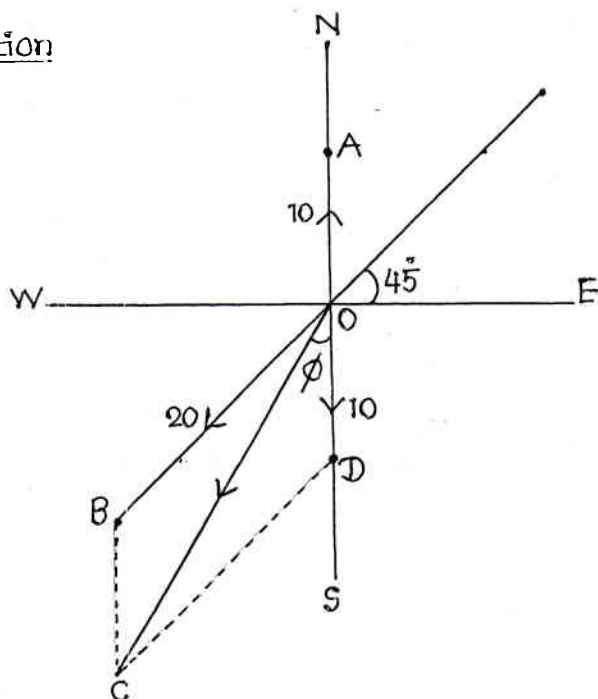
$$\theta = \tan^{-1}\left(\frac{5}{11}\right)$$

The direction in which the drops appear to fall to the person is $\tan^{-1}\left(\frac{5}{11}\right)$ east of south or $S 24.4^\circ E$.

The velocity they would hit his umbrella with 24.17 ft. per sec.

7 A steamship is travelling north at the rate of 10 m.p.h, and there is a north-east wind blowing at the rate of 20 m.p.h. In what direction will the smoke from the funnel appear to move to an observer on the ship?

Solution



$\vec{O} =$ the velocity of the smoke

relative to the observer

$\angle BOD = 45^\circ$, $OB = 20$, $OD = 10$

Let $\angle COD = \phi$

$$\tan \phi = \frac{20 \sin 45^\circ}{10 + 20 \cos 45^\circ}$$

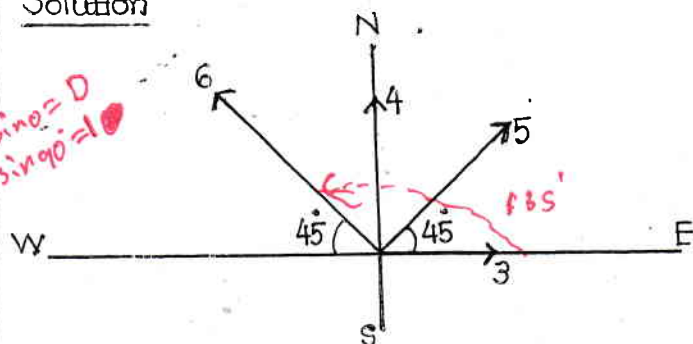
$$= \frac{20 \times \frac{\sqrt{2}}{2}}{10 + 20 \times \frac{\sqrt{2}}{2}} = \frac{10\sqrt{2}}{10 + 10\sqrt{2}} = \frac{\sqrt{2}}{1 + \sqrt{2}}$$

$$\phi = \tan^{-1} \left(\frac{\sqrt{2}}{1 + \sqrt{2}} \right) = 30.36^\circ$$

The apparent direction of the smoke is $S 30.36^\circ W$.

8 A point has velocities of 3, 5, 4 and 6 in directions east, north-east, north and north-west respectively, find the magnitude and direction of its resultant velocity.

Solution



The horizontal component,

$$X = 3 \cos 0^\circ + 5 \cos 45^\circ + 4 \cos 90^\circ + 6 \cos 135^\circ$$

$$= 3 + 5 \cos 45^\circ - 6 \cos 45^\circ$$

$$X = 3 - \frac{\sqrt{2}}{2} = 2.29$$

The vertical component,

$$Y = 3 \sin 0^\circ + 5 \sin 45^\circ + 4 \sin 90^\circ + 6 \sin 135^\circ$$

$$= 5 \sin 45^\circ + 4 + 6 \sin 45^\circ$$

$$Y = 4 + 11 \frac{\sqrt{2}}{2} = 11.78$$

The resultant velocity $V = \sqrt{X^2 + Y^2}$

$$V = \sqrt{(2.29)^2 + (11.78)^2} = 12$$

$$\tan \theta = \frac{Y}{X} = \frac{11.78}{2.29} = 5.1441$$

$\theta = \tan^{-1} 5.1441 = 79^\circ$ with horizontal

The direction of resultant makes an angle $\theta = 79^\circ$ north of east (or) $E 79^\circ N$.

Unit 8

Force, momentum, law of motion

Newton's law of motion

force = mass $\times$ acceleration

$$P = mf$$

$$(\text{pdl}) = (\text{lb})(\text{ft. per sec}^2)$$

engineers' unit = pound-weight (lb. wt.)

$$1 \text{ lb. wt.} = 32 \text{ pdl}$$

Motion with uniform acceleration

$$1. v = u + ft$$

$$2. s = ut + \frac{1}{2} ft^2$$

$$3. v^2 = u^2 + 2fs$$

u = initial velocity, v = final velocity

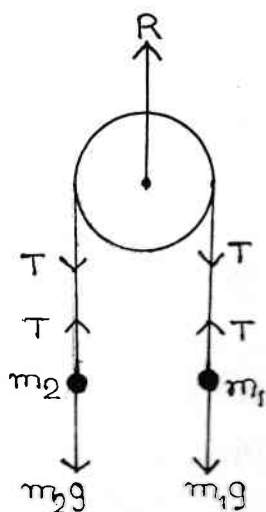
f = uniform acceleration

s = distance travelled

t = time taken

- 1 Two particles of masses m_1 and m_2 lb are connected by a light inextensible string passing over a small smooth fixed pulley. Find the resulting motion of the system and the tension in the string.

Solution



Let the tension in the string = T pdl

Suppose $m_1 > m_2$

m_1 will move downwards and m_2 will move upwards.

Let common acceleration = f ft. per sec²

The equation of motion is $P = mf$

For m_1 , $m_1g - T = m_1f$ ——— ①

For m_2 , $T - m_2g = m_2f$ ——— ②

① + ② $\Rightarrow m_1g - m_2g = m_1f + m_2f$

$(m_1 - m_2)g = (m_1 + m_2)f$

$f = \frac{(m_1 - m_2)g}{m_1 + m_2}$ ←

② $\Rightarrow T = m_2f + m_2g$

$= m_2 \frac{(m_1 - m_2)g}{m_1 + m_2} + m_2g$

$= m_2g \left(\frac{m_1 - m_2}{m_1 + m_2} + 1 \right)$

$= m_2g \left(\frac{m_1 - m_2 + m_1 + m_2}{m_1 + m_2} \right)$

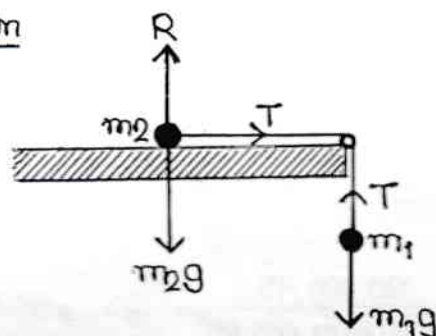
$T = \frac{2m_1m_2g}{m_1 + m_2}$ ←

The pressure on the pulley = $R = 2T$

$R = 2 \left(\frac{2m_1m_2g}{m_1 + m_2} \right) = \frac{4m_1m_2g}{m_1 + m_2}$ ←

- 2 A mass m_2 lb is placed on a smooth horizontal table, and connected by a light inextensible string passing over a small smooth pulley at the edge to a mass m_1 lb hanging freely. Find the resulting motion and the tension in the string.

Solution



Let the tension in the string = T pdl

m_1 will move downwards and m_2 will move along the table.

Let common acceleration = f ft. per sec²

The equation of motion is $P = mf$

for m_1 , $m_1g - T = m_1f$ ——— ①

for m_2 , $T = m_2f$ ——— ②

There is no motion perpendicular to the table, $R = m_2g$ ←

① + ② $\Rightarrow m_1g = m_1f + m_2f$

$m_1g = (m_1 + m_2)f$

$f = \frac{m_1g}{m_1 + m_2}$ ←

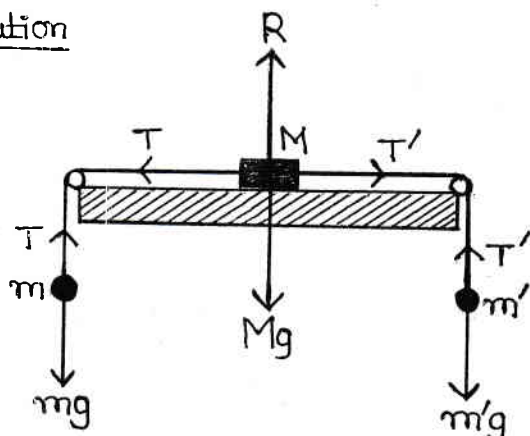
② $\Rightarrow T = \frac{m_1m_2g}{m_1 + m_2}$ ←

The pressure on the pulley = $\sqrt{T^2 + T^2}$

$= \sqrt{2T^2} = \sqrt{2} T = \sqrt{2} \frac{m_1m_2g}{m_1 + m_2}$ pdl. ←

3 A mass of M lb rest on a smooth horizontal table and is attached by two inelastic strings to masses m, m' lb ($m' > m$), which hang over smooth pulleys at opposite edges of the table. Find the acceleration of the system and the tension in the strings.

Solution



Let the tension in the strings are T and T' pdl.

The common acceleration $= f$ ft. per sec²

Suppose $m' > m$.

$\therefore m'$ will move downwards,
 m will move upwards and
 M will move along the table
 towards m' .

$$\text{for } m', \quad m'g - T' = m'f \quad \text{--- (1)}$$

$$\text{for } m, \quad T - mg = mf \quad \text{--- (2)}$$

$$\text{for } M, \quad T' - T = Mf \quad \text{--- (3)}$$

$$\text{(1) + (2) + (3)} \Rightarrow m'g - mg = m'f + mf + Mf$$

$$(m' - m)g = (m' + m + M)f$$

$$f = \frac{(m' - m)g}{m' + m + M} \quad \leftarrow$$

$$\text{(2)} \Rightarrow T = mg + mf = m(g + f)$$

$$= m \left(g + \frac{(m' - m)g}{m' + m + M} \right)$$

$$= m \left(1 + \frac{m' - m}{m' + m + M} \right) g$$

$$= m \left(\frac{m' + m + M + m' - m}{m' + m + M} \right) g$$

$$T = \frac{m(2m' + M)g}{m' + m + M} \quad \leftarrow$$

$$\text{(1)} \Rightarrow T' = m'g - m'f = m'(g - f)$$

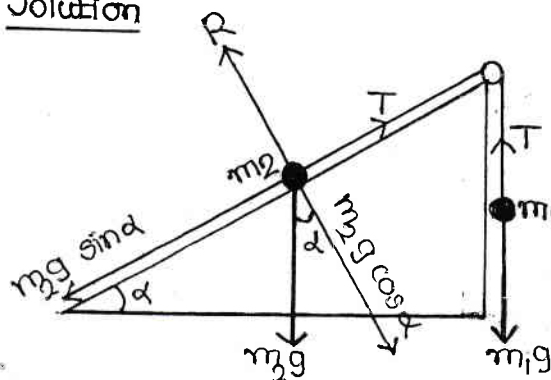
$$= m' \left(g - \frac{(m' - m)g}{m' + m + M} \right)$$

$$= m' \left(\frac{m' + m + M - m' + m}{m' + m + M} \right) g$$

$$T' = \frac{m'(2m + M)g}{m' + m + M} \quad \leftarrow$$

4 A particle of mass m_2 lb rests on the surface of a smooth plane inclined at an angle α to the horizontal, and is connected by a light inextensible string, passing over a small smooth pulley at the top of the plane, to a mass m_1 lb hanging freely. Find the resulting motion and the tension in the string.

Solution



Let the tension in the string $= T$ pdl
 the common acceleration $= f$ ft. per sec²
 If m_1 moves downward and
 m_2 moves up the plane, then

$$\text{for } m_1, \quad m_1g - T = m_1f \quad \text{--- (1)}$$

$$\text{for } m_2, \quad T - m_2g \sin \alpha = m_2f \quad \text{--- (2)}$$

There is no motion perpendicular to the plane, $R - m_2 g \cos \alpha = 0$

$$\therefore R = m_2 g \cos \alpha \leftarrow$$

$$\textcircled{1} + \textcircled{2} \Rightarrow m_1 g - m_2 g \sin \alpha = m_1 f + m_2 f$$

$$(m_1 - m_2 \sin \alpha) g = (m_1 + m_2) f$$

$$f = \frac{(m_1 - m_2 \sin \alpha) g}{m_1 + m_2} \leftarrow$$

$$\textcircled{1} \Rightarrow T = m_1 g - m_1 f = m_1 (g - f)$$

$$T = m_1 \left(g - \frac{(m_1 - m_2 \sin \alpha) g}{m_1 + m_2} \right)$$

$$= m_1 \left(\frac{m_1 + m_2 - m_1 + m_2 \sin \alpha}{m_1 + m_2} \right) g$$

$$T = \frac{m_1 m_2 (1 + \sin \alpha) g}{m_1 + m_2} \leftarrow$$

$$\text{Since } 5g \sin \alpha < 3g,$$

m_1 will move downwards and

m_2 will move up the plane.

$$\text{for } m_1, \quad m_1 g - T = m_1 f$$

$$3g - T = 3f \quad \textcircled{1}$$

$$\text{for } m_2, \quad T - m_2 g \sin \alpha = m_2 f$$

$$T - 5g \times \frac{1}{5} = 5f$$

$$T - g = 5f \quad \textcircled{2}$$

There is no motion perpendicular to the plane, $R - m_2 g \cos \alpha = 0$

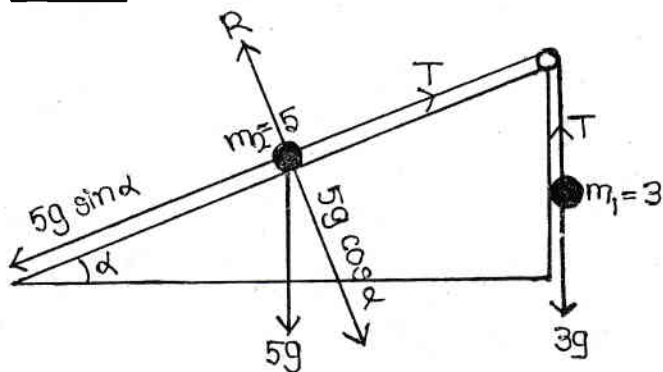
$$\textcircled{1} + \textcircled{2} \Rightarrow 2g = 8f$$

$$f = \frac{2 \times 32}{8} = 8 \text{ ft per sec}^2 \leftarrow$$

$$\textcircled{2} \Rightarrow T = 5f + g = 5 \times 8 + 32 = 72 \text{ pdl} \leftarrow$$

5 A particle, of mass 5 lb, is placed on a smooth plane whose height is 4 ft and length 20 ft. The particle is connected by a light string passing over a smooth pulley at the top of the plane to a mass of 3 lb hanging freely. Find the common acceleration and the tension of the string.

Solution



$$\sin \alpha = \frac{4}{20} = \frac{1}{5}$$

Let tension in the string = T pdl

The common acceleration = f ft. per sec^2

$$m_1 = 3 \text{ lb}, \quad m_2 = 5 \text{ lb.}$$

6 What force in pounds weight will give a mass of 9 tons a velocity of 25 m.p.h in 1 min?

Solution

$$\text{velocity} = v = 25 \text{ m.p.h}$$

$$= 25 \times \frac{5280}{60 \times 60} \text{ ft. per sec.}$$

$$v = \frac{110}{3} \text{ ft. per sec.}$$

$$\text{time} = t = 1 \text{ min} = 60 \text{ sec}$$

$$\text{initial velocity} = u = 0$$

$$v = u + ft$$

$$\frac{110}{3} = 0 + f \times 60$$

$$f = \frac{110}{3 \times 60} = \frac{11}{18} \text{ ft per sec}^2$$

$$\text{mass} = m = 9 \text{ tons} = 9 \times 2240 \text{ lb}$$

$$m = 20160 \text{ lb}$$

$$\text{The required force} = P = mf$$

$$= 20160 \times \frac{11}{18} \text{ pdl}$$

$$= 12320 \text{ pdl}$$

$$= \frac{12320}{32} \text{ lb. wt}$$

$$p = 385 \text{ lb. wt} \leftarrow$$